



**DAYANANDA SAGAR
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**Devarakaggalahalli, Harohalli, Kanakapura Road,
Dt. Bangalore South, Karnataka -562112**

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(DATA SCIENCE)

CAPSTONE PROJECT REPORT

SMART MEDICATION ADHERENCE COLLAR

2025-2026

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

(DATA SCIENCE)

SUBMITTED BY

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UNDER THE SUPERVISION OF

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Computer Science & Engineering (Data Science)

Dayananda Sagar University, Bengaluru

Academic Year: 2025–2026

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CERTIFICATE

This is to certify that the **Capstone Project** titled “**Smart Medication Adherence Collar**” has been carried out by **Alisha Ramazan Shiekh- ENG24DS0004, Bhavana V- ENG24DS0007**, Bonafide students of **Bachelor of Technology in Computer Science and Engineering (Data Science)** at the **School of Engineering, Dayananda Sagar University, Bangalore**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Data Science)** during the academic year **2025–2026**. It is further certified that all corrections and suggestions indicated during the internal assessment have been incorporated in the final report submitted to the departmental library. The project report has been approved as it satisfies the academic requirements in respect of the project work for the said degree.

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We, **Alisha Ramazan Shiekh- ENG24DS0004, Bhavana V- ENG24DS0007** are students of fourth semester B.Tech in Computer Science and Engineering (Data Science), at School of Engineering, Dayananda Sagar University, hereby declare that the Major Project titled “**Smart Medication Adherence Collar**” has been carried out by us and submitted in partial fulfilment for the award of degree in Bachelor of Technology in Computer Science and Engineering (Data Science) during the academic year **2025- 2026**

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
EV	Electric Vehicle
ESP32	Espressif Systems Processor 32-bit
IoT	Internet of Things
RTC	Real-time Clock.
BLE	Bluetooth Low Energy
GSM	Global System for Mobile Communication
SMS	Short Message Service
IMU	Inertial Measurement Unit
I2C	Inter-Integrated Circuit(communication protocol)
OLED	Organic Light-Emitting Diode
RGB	Red, Green, Blue
SSD1306	OLED Display Driver IC
WHO	World Health Organization

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ABSTRACT

Lots of people don't take their medication as they should - in fact, around 50% of those with long-term illnesses don't, and the World Health Organization says this is a huge problem all over the world. This is even more true for older adults with a lot of health issues, where about 42.6% of them aren't following their medication plans. Because of this, their health gets worse and medical care becomes more expensive. This device is a clever solution to help with this, and it's been developed to improve how much patients actually do take their medication, and so improve their health, and it doesn't need a complicated setup. It uses dependable alerts and will give out the medicine.

The intention of this project is to build a small, cheap dispenser that will both remind people to take their pills on time and reliably give them the correct amount. It will also let family or other carers know when a dose has been given. To make it work, either the user or a carer can program a timetable. A dial you can turn and a clock that is accurate in real time are used to set one or more times for doses during the day. When it is time for a dose, the system will make an audible alarm.

When it is time for a dose, the system will produce an audible alarm to alert the patient. The patient is then required to manually open the container and take the prescribed medication, as the pills are stored in bottles rather than an automated drawer system. Inside the device, a microcontroller (such as an Arduino) manages the timing and alarm functions. Additionally, a GSM/SMS module is integrated into the system, which sends a notification message to a family member or caregiver each time a dose is due. This ensures remote notification of the patient's medication events (or potential misses) even if the patient is alone or if the caregiver is out of range of Wi-Fi. The system can be powered by batteries or a USB supply for portability.

The device operates fully without requiring an Internet connection or smartphone, making it suitable for low-connectivity environments. It is based on a microcontroller system with components such as a buzzer for alerts and a GSM module for notifications. Unlike many IoT-based dispensers, all scheduling and operations are handled locally using an RTC and a manual knob interface. The system alerts users through an alarm, and medication is taken manually from stored bottles. A key feature is the GSM module, which sends an SMS notification if a dose is missed or not acknowledged. This ensures caregivers can intervene promptly, improving reliability and patient safety.

CHAPTER 1 INTRODUCTION

Medication non-adherence poses a serious public health challenge worldwide. The World Health Organization (WHO) has noted that adherence to long-term therapy among patients with chronic conditions averages only about 50%, meaning half of prescriptions may not be taken correctly. This problem is particularly acute in older adults with multiple chronic illnesses: recent literature finds roughly 42.6% of such multimorbid patients do not fully follow their medication regimens. Failure to adhere often leads to worse health outcomes, greater risk of complications, and increased healthcare costs. Addressing this gap in care, especially for vulnerable groups, is critical for preventing avoidable hospitalisations and improving long-term wellbeing.

This project focuses on a smart medicine dispenser designed for patients with complex medication schedules – especially the elderly or those with polypharmacy – who may forget doses due to memory decline or busy lives. Many older adults take multiple pills daily (for example, over 40% of Americans aged 65+ take five or more medications), making timely dosing difficult. Traditional reminder systems often assume internet connectivity or smartphone use, which may not be available for many seniors or in low-connectivity environments. Our design explicitly targets these needs: it is compact and portable for home or travel use, requires no Wi-Fi (only basic GSM cellular signals for alerts), and uses simple controls suited to users of any technical skill level. By alerting both patients and caregivers, it supports independent living while maintaining oversight.

The device's operation combines straightforward scheduling with user-activated dispensing. A real-time clock (RTC) and an easy-to-turn knob let the user or a caregiver set one or more medication times without a smartphone app. When the clock reaches a scheduled time, an audible buzzer alarm sounds. Crucially, unlike fully automated dispensers, the medication compartment (bottle or drawer) does *not* open until the user presses a release button during the alarm period. Once the alarm is triggered, pressing the button causes the actuator to open the compartment, allowing access to that dose. The user must then manually close the compartment when finished, which prevents pills from remaining exposed. All electronics (microcontroller, clock, buzzer, motor) run on low power and reset until the next dosing time. A GSM module sends an SMS notification to a designated caregiver at each scheduled event (whether the alarm is dismissed or not).

This keeps family members or healthcare providers informed even if they are not nearby or if the patient is outdoors. Because scheduling and alerts are handled on-device and via SMS, the system works entirely offline (no internet), making it reliable in remote areas.

The design rationale emphasises safety, simplicity, and autonomy. Requiring the user to open and close the compartment manually adds safety by preventing accidental or unauthorized dispensing (for example, small children could not easily trigger doses). It also gives the patient control over the process, which may improve satisfaction and compliance. From a technical standpoint, this manual-close mechanism reduces mechanical complexity and power use (the motor runs only briefly to open, not constantly managing a lid). However, it also means trade-offs: it relies on the user to respond to the alarm and to close the dispenser securely, which could slightly reduce ease-of-use for those with severe dexterity issues. Caregivers benefit from this approach because they are explicitly notified of each dosing attempt via SMS, so they can follow up if the patient ignores an alarm. The main limitation is dependence on cellular signal; in areas without mobile coverage, SMS alerts will fail. Maintenance is straightforward since there are no complex locking mechanisms to service.

The main goals for this project are:

1. **Timely Reminders** – Alert pets/patients when it's time to take medication via vibration or sound.
2. **Dosage Tracking** – Record and log each dose administered to prevent missed or double doses.
3. **Real-Time Monitoring** – Continuously monitor medication adherence status throughout the day.
4. **Mobile App Integration** – Sync collar data with a smartphone app for easy management.
5. **Caregiver Alerts** – Notify caregivers or owners instantly if a dose is missed.
6. **Medication Schedule Setup** – Allow users to program custom medication schedules easily.

7. **Compliance Reporting** – Generate weekly/monthly adherence reports for doctors or vets.
8. **Battery Efficiency** – Ensure long battery life for uninterrupted, 24/7 operation.
9. **Comfortable Wearable Design** – Design a lightweight, skin-friendly collar for continuous wear.
10. **Data Security** – Protect all health and medication data with secure encryption.

In summary, this smart dispenser addresses the high rate of missed medication doses in vulnerable populations by combining reliable alarms with a controlled dispensing mechanism. In doing so, it aligns with evidence that better adherence yields better health outcomes. The following sections will detail its technical implementation and expected impact on adherence and patient safety.

CHAPTER 2 PROBLEM DEFINITION

Managing medication schedules is a critical yet challenging task for patients, elderly individuals, and their caregivers. Many patients suffering from chronic illnesses, post-surgical recovery, or age-related conditions require strict and timely medication routines. However, forgetfulness, cognitive decline, and busy daily schedules often lead to missed or incorrect doses, resulting in serious health deterioration.

Currently, dependent patients — especially the elderly and those with memory-related disorders such as Alzheimer's or dementia — lack a reliable, automated solution that actively monitors and enforces medication adherence. Existing solutions such as pill boxes, phone reminders, or manual caregiver supervision are inconsistent, non-automated, and provide no real-time tracking or instant alert system.

Family members and healthcare professionals often have no way of knowing whether a patient has taken their medication on time, leading to gaps in care and potential medical emergencies. There is an urgent need for a smart device that attaches directly to the medication bottle, automatically reminds patients to take their medication at the right time, tracks whether the bottle was opened or the dose was taken, notifies caregivers in case of missed doses, and generates adherence reports for medical review. The **Smart Medication Adherence Collar** bridges this critical gap through IoT-based monitoring, real-time alerts, and mobile connectivity.

This now clearly focuses on:

- **Target users** – humans, elderly, chronic illness patients
- **Key conditions** – Alzheimer's, dementia, post-surgical recovery
- **Problem** – missed doses, no real-time tracking
- **Solution** – smart collar that attaches directly to the medication bottle

CHAPTER 3 LITERATURE SURVEY

Medication non-adherence remains one of the most serious problems in modern healthcare management, particularly for patients suffering from chronic illnesses. Non-adherence to prescribed medication can lead to adverse health outcomes, increased hospitalizations, and additional healthcare costs. Research consistently confirms that improving medication adherence may have a more profound impact on public health than developing new therapies, and effective solutions require a holistic approach that integrates reliable measurement, patient-centered care, technological innovation, and system-level support. These findings strongly motivated the development of automated systems like the Smart Medication Adherence Collar to bridge the gap between prescribed treatment and real-world patient behavior.

Several researchers have explored IoT-based solutions to address the medication adherence problem. A novel IoT-based Medication Adherence System was proposed to combat the pervasive issue of non-adherence among chronic disease patients, leveraging real-time monitoring and timely reminders to improve medication intake and patient outcomes (ScienceDirect, 2024). Similarly, Islam et al. designed an IoT-based Digital Medicine Box using a NodeMCU ESP8266 microcontroller that provides reminders at designated times, reduces the burden of medication management, and uses emergency switches to trigger alert SMS messages to caretakers and doctors (EJECE, 2023). These systems demonstrate the viability of integrating IoT hardware and software components for automated medication tracking, directly paralleling the architecture used in this project.

The Real-Time Clock (RTC) module is a fundamental component in time-based medication reminder systems. The DS3231 RTC module is widely used for data logging, timekeeping, time stamping, and alarms in medication systems. An RTC interface always keeps track of real time irrespective of the power supply, and if the main power goes out, the RTC module draws power automatically from a coin-cell battery, ensuring the microcontroller gets the correct time without error upon restart (PMC, 2024). Research by Philip et al. further demonstrated a smart medicine box using an Arduino UNO, NodeMCU, RTC module, and servo motor, where the RTC-controlled system opens the medicine box at scheduled times to ensure timely medication intake and reminders (EJENG, 2025). These findings validate the use of an RTC module in the proposed collar system for accurate schedule-based triggering.

The use of buzzer and LED-based alert systems has been widely adopted as an effective and low-cost reminder mechanism. Based on an RTC interfaced to a microcontroller, the programmed time and number of pills for a medicine is displayed on an LCD along with a buzzer sound to alert the patient about taking the appropriate medicine. If the patient does not press the done button, a reminder is sent via SMS on their cell phone using a GSM modem (Industrial Engineering Journal, 2023). This multi-level alerting approach — combining physical alerts with digital notifications — is a design principle that the Smart Medication Adherence Collar also adopts through its buzzer, LED, and mobile app notification system.

Detection of bottle opening using sensors is a critical step in verifying actual medication intake. A two-step system was proposed for detecting when a pill bottle was opened using commercial wireless smart-bottle technologies with Bluetooth transmission, and when a pill was consumed using a custom-designed wireless smart necklace with a piezoelectric sensor. These two mechanisms with the mobile application can passively monitor adherence and inform caregivers of patient status (ResearchGate, 2016). Furthermore, Hall effect sensors are solid-state devices not prone to mechanical wear, capable of operating at much higher speeds than mechanical sensors, and are immune to dust, dirt, mud, and water, making them better suited for position sensing than optical and electromechanical methods (Wikipedia). These findings support the use of a Hall effect or reed switch sensor in this project to detect bottle cap opening reliably and efficiently.

Wearable devices have emerged as a powerful platform for continuous medication adherence monitoring. A wearable sensor system was designed for medication adherence prediction, introducing a two-step system for identifying the opening of a pill container using smart bottle technologies and the consumption of a pill using a smart necklace with a piezoelectric sensor, with a mobile application automatically monitoring adherence and providing caregivers with information regarding patient status (ResearchGate, 2016). A scoping review published in JMIR Human Factors further confirmed that wearable devices are useful interfaces for recording patient adherence, though most studies used them primarily for notification or recording functions without fully leveraging the multiple sensors embedded in wearable devices (JMIR, 2024). This gap highlights the novelty and potential of the Smart Medication Adherence Collar, which integrates sensing, alerting, and reporting in one unified wearable system.

Cloud platforms play a vital role in centralizing and analyzing patient health data. Liu et al. (2020) highlighted the role of cloud computing in healthcare systems, where data from IoT devices is uploaded to cloud platforms for processing and analysis. The integration of cloud-based systems allows for centralized monitoring of multiple patients, facilitating more efficient management of patient care. Moreover, cloud computing allows for the use of machine learning algorithms to analyze trends in health data and predict potential health issues, enabling a personalized healthcare approach (IJSRA, 2025). The Smart Medication Adherence Collar adopts this cloud architecture by uploading timestamped dose data for compliance report generation and caregiver access.

CHAPTER 4

PROJECT DESCRIPTION

Smart Medication Adherence Collar — System Workflow

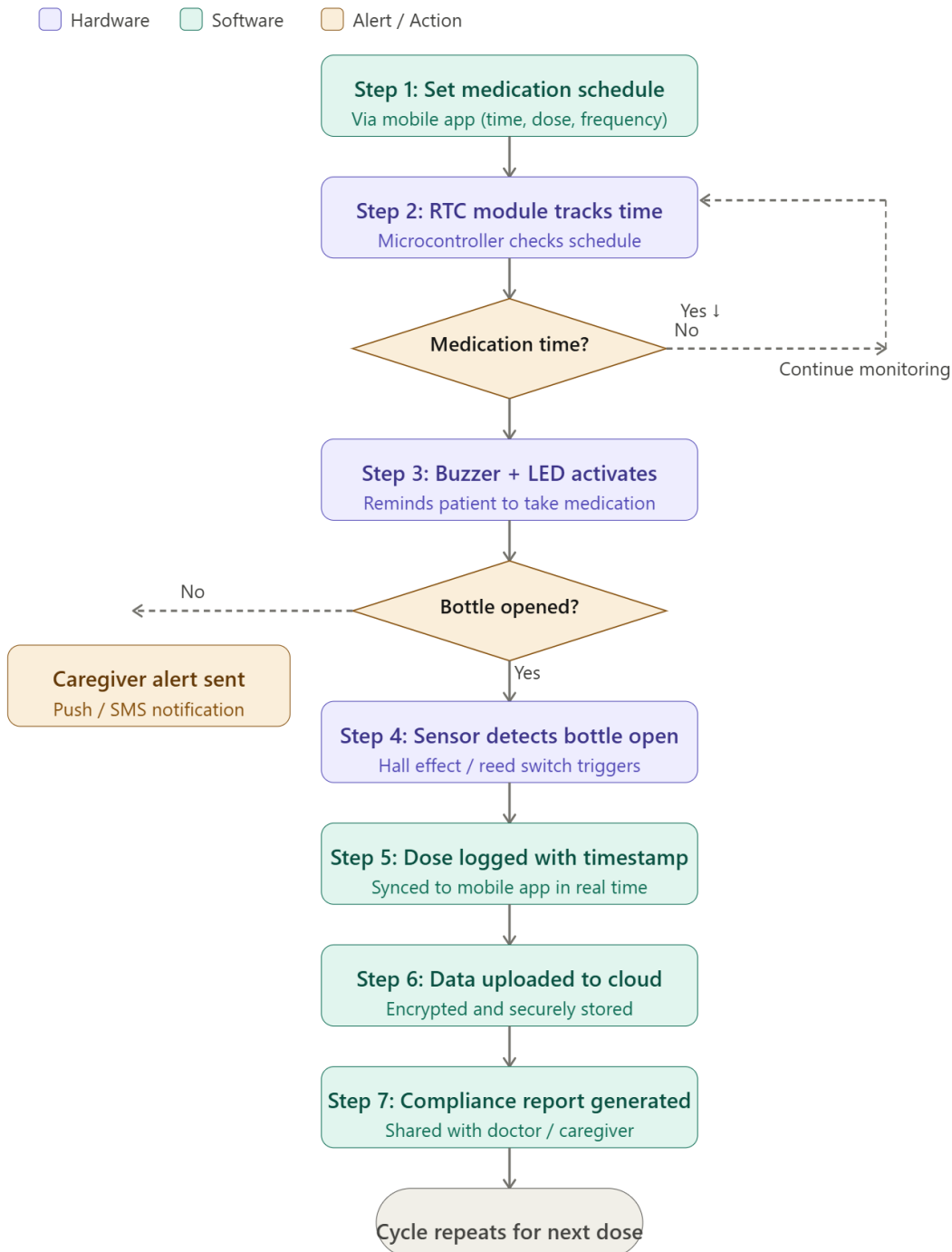


Fig: Smart Medication Adherence Collar – System Workflow

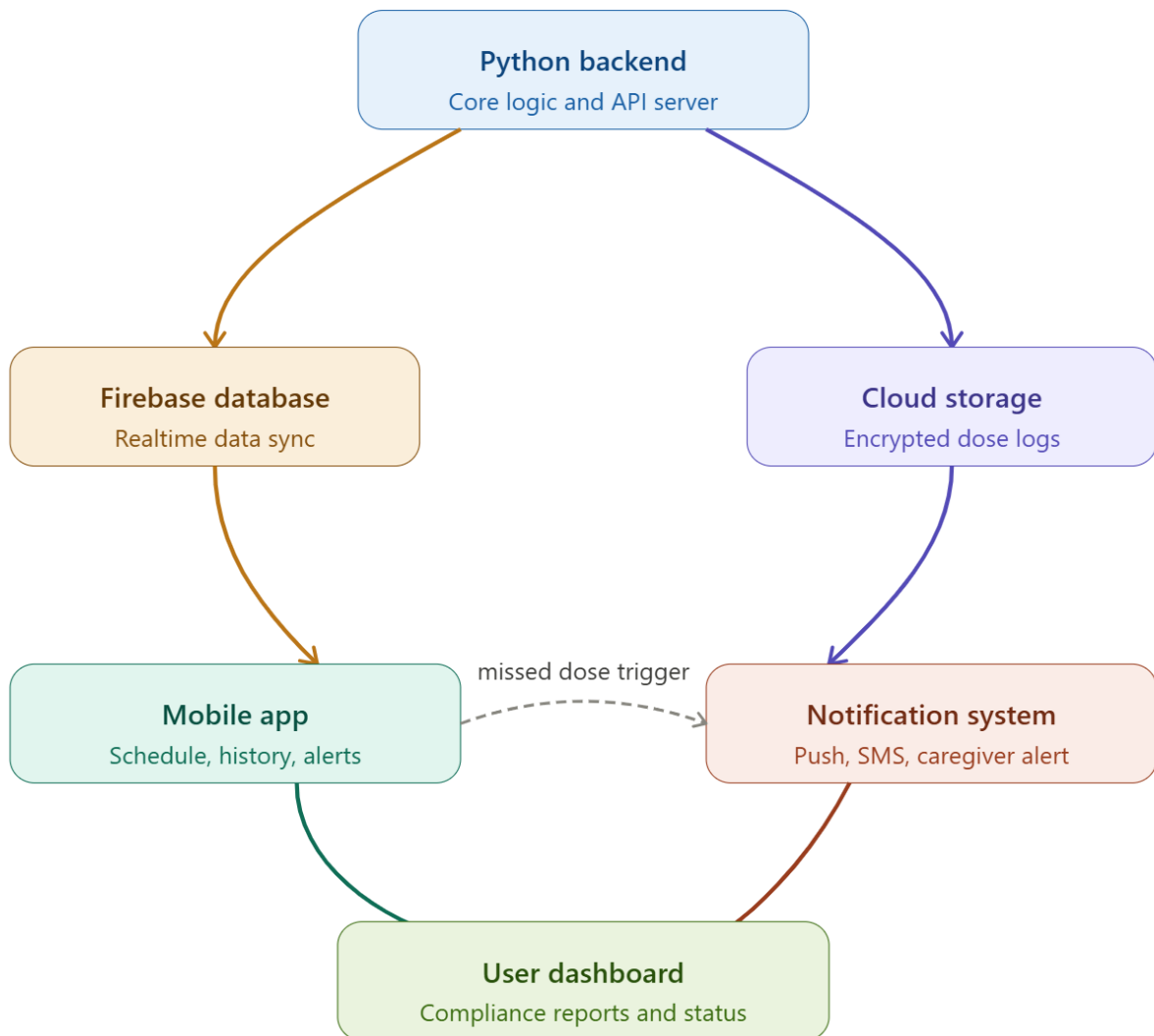


Fig 4.1.2 Software Methodology Workflow

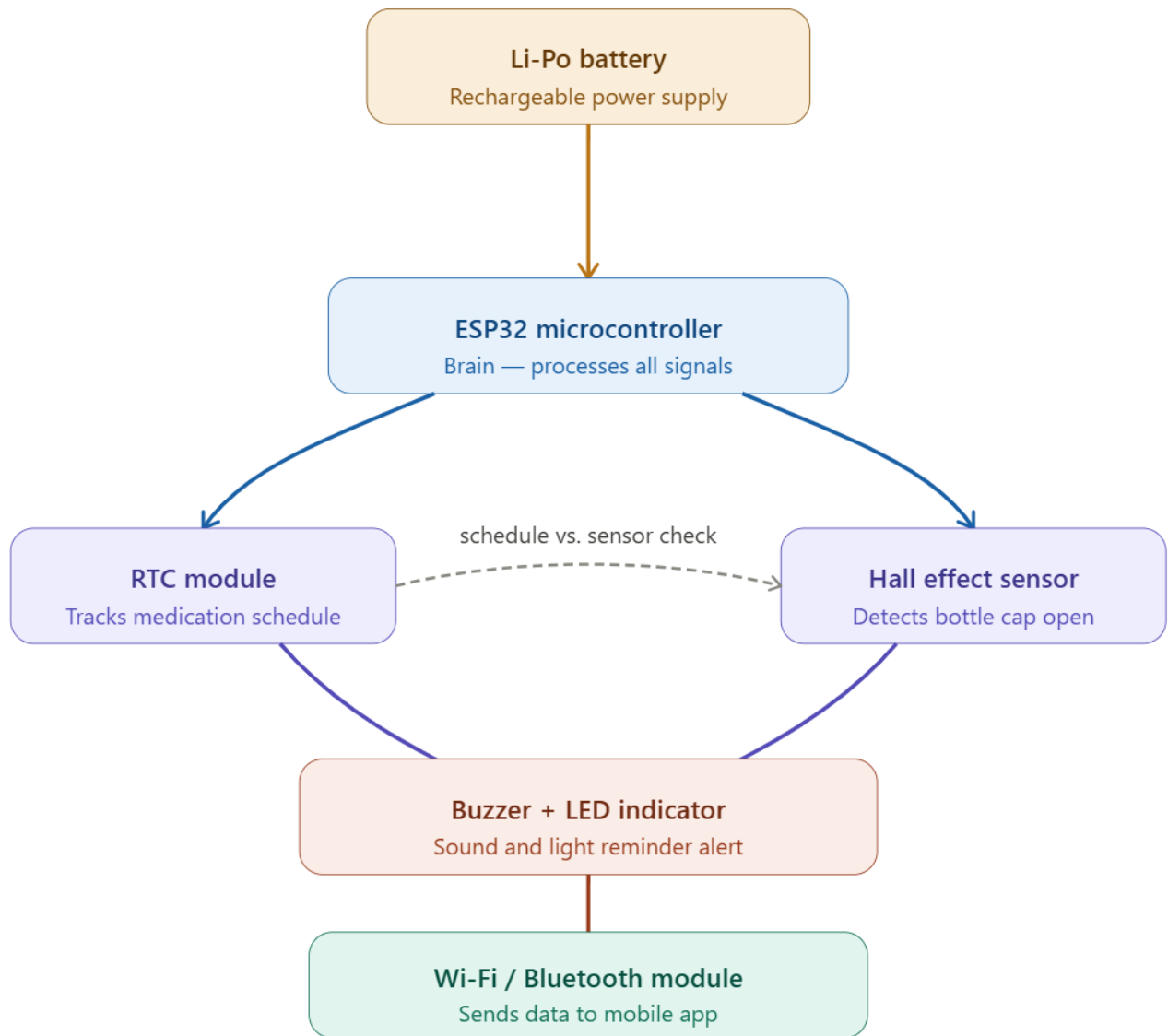


Fig 4.1.3 Hardware Methodology Work Flow

This unified methodology integrates all hardware and software components into one cohesive system which functions as a self-sustaining Smart Medication Adherence Collar that automatically monitors, reminds, and logs patient medication intake in real time.

1. **Medication Schedule Setup** The patient or caregiver sets the medication schedule (time, dose, and frequency) through a mobile app.
2. **Real-Time Clock (RTC) Monitoring** The RTC module on the collar continuously tracks time, and the microcontroller regularly checks whether it matches the set medication schedule.
3. **Medication Time Check** The system checks: *Is it medication time?*
4. **No** → The system continues monitoring in a loop until the scheduled time arrives.
5. **Yes** → Proceeds to the next step.

4. **Alert Activation** Once medication time is reached, the collar activates a **buzzer and LED light** to remind the patient to take their medication.
5. **Bottle Opened Check** The system checks: *Has the medicine bottle been opened?*
 - **No** → A **caregiver alert is sent** via Push Notification or SMS to inform them that the dose was missed.
 - **Yes** → Proceeds to the next step.
6. **Sensor Detection** A **Hall effect sensor or reed switch** on the bottle detects that the cap has been opened and triggers the logging process.
7. **Dose Logged with Timestamp** The dose intake is automatically recorded with the exact date and time, and this data is **synced to the mobile app in real time**.
8. **Cloud Upload** The logged data is **uploaded to the cloud**, where it is encrypted and securely stored for future reference.
9. **Compliance Report Generation** A **compliance report** is automatically generated from the stored data and shared with the patient's doctor or caregiver for monitoring.
10. **Cycle Repeats** The entire workflow **resets and repeats** for the next scheduled dose, ensuring continuous medication adherence tracking.

The closed-loop system ensures medication adherence through its data-based control mechanism which autonomously detects missed doses and alerts caregivers in real time. The system components work together to achieve accurate dose tracking while sustaining consistent patient compliance monitoring.

4.1 ASSUMPTIONS AND DEPENDENCIES

- **Schedule Setup** It is assumed that the patient or caregiver will correctly set the medication schedule including time, dose, and frequency through the mobile app before the system begins operation. The entire workflow is dependent on this initial setup being accurate and complete.
- **Device Usage** It is assumed that the patient will wear the collar and keep the medicine bottle nearby at all times. Opening the medicine bottle is also assumed to be directly equivalent to the patient actually consuming the prescribed medication.
- **Hardware Functionality** It is assumed that the microcontroller and RTC module will function continuously without any power failure or hardware malfunction. All core system operations such as time checking, alert triggering, and dose logging are fully dependent on the microcontroller running without interruption.

- **Sensor Detection** It is assumed that the Hall effect or reed switch sensor is sensitive and accurate enough to detect every instance of the bottle being opened. Dose detection and logging are entirely dependent on this sensor triggering correctly each time the bottle cap is removed.
- **Alert System** It is assumed that the buzzer and LED alert system is sufficiently loud and visible to effectively remind the patient when it is time to take their medication. Caregiver alerts for missed doses are dependent on a third-party push notification or SMS service being active and functional at all times.
- **Connectivity** It is assumed that the patient's device maintains a stable internet or Bluetooth connection at all times. Real-time data syncing to the mobile app and cloud upload are both fully dependent on this wireless connection being consistently available.
- **Mobile Application** It is assumed that the mobile app is installed and actively running on the caregiver's or patient's smartphone throughout the day. The entire system depends on the mobile app for schedule management, real-time notifications, and compliance report viewing.
- **Cloud and Data Security** It is assumed that the cloud platform is always available and accessible for storing and retrieving medication compliance data. Secure data storage is dependent on a reliable encryption protocol being in place to protect sensitive patient health information at all times.
- **Compliance Reporting** It is assumed that the generated compliance reports are shared with and reviewed by the designated doctor or caregiver regularly. The report sharing feature is dependent on the caregiver or doctor having active access to the mobile app or cloud dashboard.
- **Power Supply** The entire Smart Medication Adherence Collar system is dependent on a consistent and adequate power source such as a rechargeable battery. It is assumed that the battery will be regularly charged to ensure uninterrupted system operation throughout the day.

CHAPTER 5 Requirments

HARDWARE AND FUNCTIONAL REQUIREMENTS

1. ESP32-C3 Supermini (with USB Verification Port)

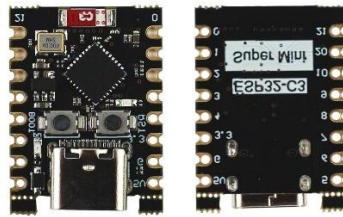


Fig 5.1: ESP32-C3 Supermini (with USB Verification Port)

The ESP32-C3 Supermini is a compact, low-power microcontroller development board based on the ESP32-C3 chip by Espressif Systems. It is a RISC-V single-core processor with built-in Wi-Fi and Bluetooth Low Energy (BLE) connectivity, making it ideal for IoT-based embedded systems.

In the Smart Medication Adherence Collar, the ESP32-C3 Supermini acts as the **central brain of the entire system**. It continuously runs the main program loop, checking the medication schedule set via the mobile app against the current time provided by the RTC module. When medication time is reached, it triggers the buzzer and LED alert. It also reads the input from the Hall effect or reed switch sensor to detect whether the medicine bottle has been opened. Based on this reading, it either logs the dose with a timestamp or sends a missed dose alert to the caregiver via Wi-Fi. All collected data is then transmitted to the cloud platform for secure storage and compliance report generation.

The **USB Type-C verification port** on the Supermini serves a critical role — it is used during development and testing to upload the firmware, monitor serial output for debugging, and verify that the microcontroller is functioning correctly before deployment on the collar.

2. MPU6050 Module (6-pinbreakout board)



Fig 5.2: MPU6050 Module (6-pinbreakout board)

The MPU6050 is a 6-axis Inertial Measurement Unit (IMU) sensor combining a 3-axis accelerometer and a 3-axis gyroscope on a single chip by InvenSense. It communicates via I2C protocol and is commonly used in wearable and motion-sensing applications.

In the Smart Medication Adherence Collar, the MPU6050 is used for motion and orientation detection. It detects patient movement to monitor whether the collar is being worn, identifies sudden acceleration changes for fall detection, and triggers an emergency caregiver alert if a fall is detected. Motion data is synced to the cloud alongside medication logs for a complete patient health overview. It communicates with the ESP32-C3 Supermini via the SDA and SCL pins using the I2C protocol.

3.A3144 hall effect sensor module

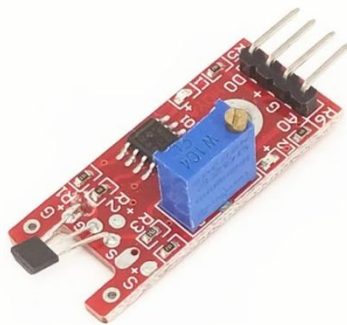


Fig 5.3: A3144 hall effect sensor module

The A3144 is a digital Hall effect sensor that detects the presence of a magnetic field and outputs a digital signal. It is a non-contact, solid-state sensor widely used for proximity and position detection in embedded systems.

In the Smart Medication Adherence Collar, the A3144 Hall Effect Sensor is used to **detect when the medicine bottle cap is opened or closed**. A small magnet is attached to the bottle cap, and the sensor is fixed near the bottle. When the cap is closed the magnet is near the sensor and it outputs LOW. When the patient opens the cap the magnet moves away, the sensor outputs HIGH, and this signal is sent to the ESP32-C3 Supermini which logs the dose with a timestamp and syncs it to the mobile app in real time.

4.OLED ‘Display Module’



Fig 5.4 **OLED ‘Display Module’**

The OLED 0.96 inch display is a small, high-contrast self-lit display module that does not require a backlight. It uses the SSD1306 driver chip and communicates via I2C protocol, making it popular in compact embedded and wearable systems.

In the Smart Medication Adherence Collar, the OLED 0.96" display acts as the **local visual interface for the patient**. It displays the current time and date fetched from the RTC module, upcoming medication schedule, dose confirmation messages when the bottle is opened, and missed dose warnings. This allows the patient to quickly check their medication status directly on the collar without needing to open the mobile app every time.

5.RGB LED Module



Fig: 5.5 **RGB LED Module**

An RGB LED is a single LED module that combines Red, Green, and Blue light elements, capable of producing a wide range of colors by varying the intensity of each channel. It is commonly used as a visual indicator in embedded and wearable systems.

In the Smart Medication Adherence Collar, the RGB LED acts as a **color-coded visual alert indicator** for the patient. Each color represents a different system status — **Green** indicates the dose was taken successfully, **Red** indicates a missed dose or caregiver alert has been triggered, and **Blue** indicates the system is active and monitoring. This gives the patient and nearby caregivers an instant visual status of the medication adherence without needing to check the OLED display or mobile app.

6.3.3V Buzzer



Fig:5.6 3.3V Buzzer

A 3.3V buzzer is a small electromechanical or piezoelectric sound-producing component that generates an audible beep or tone when powered. It is widely used in embedded systems as an audio alert or notification device.

In the Smart Medication Adherence Collar, the 3.3V buzzer serves as the **primary audio alert mechanism** for the patient. When the scheduled medication time is reached, the ESP32-C3 Supermini triggers the buzzer to produce an audible beep to remind the patient to take their medicine. If the bottle is not opened within the set time window, the buzzer sounds again as a second reminder before a missed dose alert is sent to the caregiver. Being 3.3V compatible it directly interfaces with the ESP32-C3 Supermini without needing any additional voltage regulation components.

7.6 x 6mm tactile button



Fig 5.7 6 x 6mm tactile button

A 6x6mm tactile push button is a small momentary switch that completes or breaks an electrical circuit when pressed and released. It is one of the most commonly used input components in embedded systems for user interaction.

In the Smart Medication Adherence Collar, the 6x6mm tactile button serves as the **manual user input control**. The patient can press it to acknowledge a medication reminder and stop the buzzer alert, confirm that the dose has been taken, or navigate through information displayed on the OLED screen such as schedule and dose history. It provides a simple and physical way for the patient to interact with the collar system without relying solely on the mobile app.

8. Breadboard

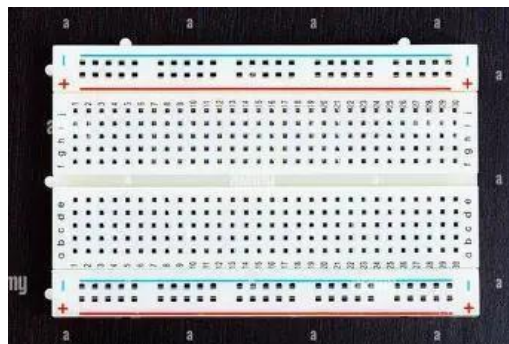


Fig 5.8. Breadboard

A breadboard is a solderless prototyping board used to build and test electronic circuits temporarily. It allows components and wires to be inserted and removed easily without any soldering, making it ideal for prototyping and testing embedded systems.

In the Smart Medication Adherence Collar, the breadboard is used during the **prototyping and testing phase** of the project. All hardware components including the ESP32-C3 Supermini, MPU6050, A3144 Hall Effect Sensor, OLED display, RGB LED, buzzer, and tactile button are connected and tested on the breadboard before being transferred to a final PCB or permanent circuit. It allows quick modifications, easy debugging, and circuit verification without any permanent connections.

9.TP4056 with type C



Fig 5.9 TP4056 with type C

The TP4056 is a single-cell lithium-ion and lithium polymer battery charging module. The Type-C variant uses a USB Type-C port for charging input, making it modern, convenient, and widely compatible with current charging cables and adapters.

In the Smart Medication Adherence Collar, the TP4056 module serves as the **battery charging and protection unit**. It safely charges the 250mAh LiPo battery via the USB Type-C port when connected to a power source. It automatically stops charging when the battery reaches full capacity, preventing overcharging. The built-in protection circuit also safeguards the battery from over-discharge and short circuits, ensuring long battery life and safe operation of the wearable collar at all times.

9.270mAh battery



Fig 5.10 270mAh battery

A 270mAh Lithium Polymer (LiPo) battery is a slightly higher capacity compact rechargeable battery compared to the 250mAh variant. It offers a better energy reserve while maintaining a lightweight and slim form factor, making it well suited for wearable embedded systems.

In the Smart Medication Adherence Collar, the 270mAh LiPo battery acts as a **backup or up-graded power source** alongside or in place of the 250mAh battery. The slightly higher capacity provides extended operating time for the ESP32-C3 Supermini, sensors, OLED display, buzzer, and RGB LED, ensuring the collar runs for longer durations between charges. It connects to the TP4056 charging module via the JST connector for safe and easy recharging through the USB Type-C port.

10.Velcro Straps



Fig 5.11 Velcro Straps

Velcro straps are fastening strips made of two interlocking nylon fabric surfaces — a hook side and a loop side — that stick together when pressed and can be separated easily by peeling apart. They are widely used in wearable devices for secure and adjustable attachment.

In the Smart Medication Adherence Collar, the Velcro straps serve as the **physical wearable attachment mechanism**. They are used to securely fasten the collar housing — containing the ESP32-C3 Supermini, sensors, battery, OLED display, and other components — around the patient's neck or wrist comfortably. The adjustable nature of Velcro straps ensures the collar fits patients of different sizes, and the easy peel-and-stick mechanism allows the patient to put on and remove the collar effortlessly without any tools or clasps.

11.Jumper Wires



Fig.5.12.Jumper wires

Jumper wires are flexible insulated electrical wires with connector pins at both ends used to establish connections between components on a breadboard or between modules and a microcontroller. They are available in Male-to-Male, Male-to-Female, and Female-to-Female configurations.

In the Smart Medication Adherence Collar, jumper wires are used during the **prototyping and testing phase** to establish all electrical connections between components on the breadboard. They connect the ESP32-C3 Supermini to the MPU6050, A3144 Hall Effect Sensor, OLED display, RGB LED, buzzer, and tactile button. Male-to-Female wires are used to connect module pins to the breadboard, while Male-to-Male wires link components across the breadboard rails, allowing quick and easy circuit modifications during development and testing.

12.Resistors

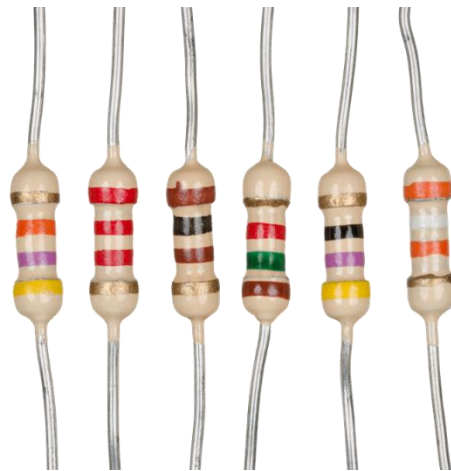


Fig:5.13 Resistors

A resistor is a passive two-terminal electronic component that limits or regulates the flow of electrical current in a circuit. They are one of the most fundamental and commonly used components in any electronic circuit design.

In the Smart Medication Adherence Collar, resistors are used at several points in the circuit for **current limiting and voltage protection**. They are connected in series with the RGB LED to prevent excess current from damaging the LED. Pull-up or pull-down resistors are used with the tactile button and Hall effect sensor to ensure stable and noise-free digital signal readings at the ESP32-C3 Supermini GPIO pins. They also protect sensitive components like the OLED display and buzzer from voltage spikes, ensuring reliable and safe operation of the entire system.

CHAPTER 6 METHODOLOGY

Data Acquisition Layer

The data acquisition layer forms the foundation of the Smart Medication Adherence Collar system. The ESP32-C3 Supermini continuously reads inputs from all connected hardware sensors in real time. The A3144 Hall Effect Sensor monitors the medicine bottle cap for open or closed status, the MPU6050 captures motion and orientation data to detect patient activity and falls, and the tactile button captures manual user acknowledgements. All raw sensor readings are collected at regular intervals by the microcontroller and passed to the next processing layer.

2. Preprocessing Layer

Once raw sensor data is collected, it undergoes preprocessing within the ESP32-C3 Supermini before any action is taken. The Hall effect sensor output is filtered to eliminate noise and debounce false triggers caused by vibration or interference. The MPU6050 accelerometer and gyroscope data is smoothed using a basic moving average filter to remove unnecessary motion spikes. The RTC module timestamp is appended to every data reading to ensure accurate time-based logging. Only valid and clean data is passed forward for decision making and transmission.

3. Decision and Alert Layer

After preprocessing, the microcontroller runs a real-time decision logic loop. It compares the current time from the RTC module against the stored medication schedule. If the scheduled time is reached, the buzzer and RGB LED are activated to alert the patient. The system then monitors the Hall effect sensor for a bottle open event within a defined time window. If the bottle is opened the dose is marked as taken and logged with a timestamp. If the bottle is not opened within the window the system triggers a missed dose alert and sends a push notification or SMS to the caregiver through Wi-Fi.

4. Data Communication Layer

The data communication layer handles all wireless transmission of medication and sensor data using the ESP32-C3 Supermini's built-in Wi-Fi (802.11 b/g/n). The microcontroller connects to the local Wi-Fi network and transmits timestamped dose logs, missed dose alerts, and motion data to the cloud platform in real time. Data is sent using lightweight communication protocols such as MQTT or HTTP REST API to ensure fast and reliable transmission even on low bandwidth networks. The mobile application receives this data instantly and displays it to the patient and caregiver in a readable format.

5. Cloud Storage and Data Management Layer

All data transmitted from the collar is received and stored on a cloud platform such as Firebase or AWS IoT. The data is encrypted during transmission using SSL/TLS protocols and stored securely in the cloud database. Each entry includes a unique patient ID, timestamp, dose status (taken or missed), and motion activity data. The cloud platform organizes this data chronologically and makes it accessible to authorized users such as doctors and caregivers through the mobile application dashboard.

6. AI Anomaly Detection Layer (Proposed Future Enhancement)

As a future enhancement to the system, an AI-based anomaly detection layer is proposed to further improve patient safety and medication management. A lightweight machine learning model such as an Isolation Forest or LSTM-based neural network would be trained on historical dose logs and motion data to identify abnormal patterns in patient behavior. For example if a patient who regularly takes medication at a fixed time suddenly shows multiple consecutive missed doses, or if the MPU6050 detects an unusual fall-like motion pattern, the AI model would flag this as an anomaly and trigger a priority alert to the caregiver or doctor. This layer would run on the cloud platform, analyzing incoming data streams and continuously improving its predictions over time as more patient data is collected.

7. Compliance Report Generation Layer

The final layer of the methodology involves automated compliance report generation. The cloud platform periodically compiles all stored dose logs, missed dose events, and motion activity data into a structured compliance report for each patient. This report includes a daily and weekly summary of medication adherence rates, timestamps of each dose taken or missed, and any anomaly alerts triggered. The report is automatically shared with the designated doctor or caregiver through the mobile application, enabling data-driven clinical decisions and timely medical intervention when required.

CHAPTER 7

IMPLEMENTATION

Arduino code:

1. Libraries and database url reading

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <Wire.h>
#include <Adafruit_SSD1306.h>
#include <Preferences.h>

#define HALL_PIN 2
#define LED_R 3
#define LED_G 4
#define LED_B 5
#define BUZZER_PIN 6
#define BUTTON_PIN 7
#define SDA_PIN 8
#define SCL_PIN 9
#define MPU_ADDR 0x68
```

1. Dose detection logic

```
if (hall == 0 && !lidOpen) {
    lidOpen = true;
    movementDetected = false;
    lidOpenTime = millis();

    baseAx = ax;
    baseAy = ay;
    baseAz = az;
}

if (lidOpen && !movementDetected) {
    int dx = abs(ax - baseAx);
```



```

int dy = abs(ay - baseAy);
int dz = abs(az - baseAz);

if (dx > movementDeltaThreshold ||
    dy > movementDeltaThreshold ||
    dz > movementDeltaThreshold) {
    movementConfirmCount++;
}

if (movementConfirmCount >= movementConfirmNeeded) {
    movementDetected = true;
}
}

if (hall == 1 && lidOpen && movementDetected) {
    registerDoseEvent();
}

```

2. Firebase upload

```

bool firebasePutJson(const String& path, const String& json) {
    if (WiFi.status() != WL_CONNECTED) return false;

    HTTPClient http;
    http.begin(makeUrl(path));
    http.setTimeout(1500);
    http.addHeader("Content-Type", "application/json");

    int code = http.PUT(json);
    http.end();

    return (code >= 200 && code < 300);
}

```

3. Offline queue

```
void enqueueOfflineDoseEvent(unsigned long epochVal,
    const String& timeText,
    int pillsRemaining,
    int dosesVal) {
    int count = prefs.getInt("qCount", 0);

    String payload = String(epochVal) + "|" +
        timeText + "|" +
        String(pillsRemaining) + "|" +
        String(dosesVal);

    prefs.putString(("q_" + String(count)).c_str(), payload);
    prefs.putInt("qCount", count + 1);
}
```

4. Firebase initialization

```
Future<void> main() async {
    WidgetsFlutterBinding.ensureInitialized();
    await Firebase.initializeApp();
    runApp(const MyApp());
}
```

5. Realtime Listener

```
StreamBuilder<DatabaseEvent>(
    stream: FirebaseDatabase.instance.ref("device").onValue,
    builder: (context, snapshot) {
        if (!snapshot.hasData || snapshot.data!.snapshot.value == null) {
            return const Center(child: Text("No Data Found"));
        }
    },
```

```

}

final data = snapshot.data!.snapshot.value as Map;

return Text("Pills Left: ${data['currentPills']}");
},
)

```

6. History listener

```

StreamBuilder<DatabaseEvent>(
  stream: FirebaseDatabase.instance.ref("history").onValue,
  builder: (context, snapshot) {
    if (!snapshot.hasData || snapshot.data!.snapshot.value == null) {
      return const Text("No history available");
    }

    final history = snapshot.data!.snapshot.value as Map;
    return ListView(
      children: history.entries.map((entry) {
        final item = entry.value as Map;
        return ListTile(
          title: Text(item['event']),
          subtitle: Text(item['time']),
        );
      }).toList(),
    );
  },
)

```

7. Firebase structure

```

{
  "device": {
    "status": "online",
    "currentPills": 56,
    "totalDoses": 26,

```

```
"lastDoseText": "2026-04-13 15:56:22",  
"lidOpen": 0,  
"movementDetected": 0,  
"missedDoseAlert": 0,  
"offlineQueue": 0  
  
},  
"history": {  
  "1776089018": {  
    "event": "dose_taken",  
    "time": "2026-04-13 15:56:22",  
    "remainingPills": 56,  
    "totalDoses": 26  
  }  
}  
}
```

CHAPTER 8 EXPERIMENTATION

The project **Smart Medication Tracker Pod with IoT-Based Monitoring and Offline Synchronization** was successfully implemented and tested using a working prototype. The system demonstrated reliable medication adherence tracking, real-time cloud synchronization, and offline data handling. The following system performances and outcomes were achieved:

8.1 BACKEND

1. Functiona Smart Medication Tracking System

The developed system successfully detects medication intake using a dual-sensor mechanism and ensures accurate adherence monitoring. The system integrates:

- **Hall Sensor** to detect lid opening and closing events
- **MPU6050 Accelerometer** to detect movement during pill dispensing

The combined sensor logic ensured that a dose is registered only when both lid activity and movement are detected, eliminating false positives caused by minor disturbances.

The system maintained consistent and accurate detection of dose events without manual intervention

2. IoT-Based Monitoring and Control System

An IoT-based monitoring system was implemented using the ESP32 microcontroller, enabling real-time communication with the Firebase Realtime Database. The system effectively:

- Collected real-time data such as pill count, movement, lid status, and dose events
- Updated cloud database continuously via Wi-Fi
- Triggered alerts using buzzer and LED for missed doses
- Displayed live system data on an OLED screen

The system operated autonomously and ensured reliability and safety without requiring continuous user interaction.

```
{
  "ax": 14032,
  "ay": -6252,
  "az": -292,
  "button": 1,
  "currentPills": 25,
  "doseTaken": 0,
  "hall": 0,
  "lastDoseEpoch": 1776098157,
  "lastDoseText": "2026-04-13 16:35:57",
  "lidOpen": 1,
  "missedDoseAlert": 1,
  "movementDetected": 1,
  "offlineQueue": 0,
  "status": "online",
  "totalDoses": 35
}
```

Fig 8.1 Real-time JSON data from Firebase during system operation

3. Offline Data Handling and Synchronization

The system was designed to function even in the absence of internet connectivity. During offline conditions:

- Dose events were stored locally using ESP32 memory (Preferences)
- A queue mechanism was implemented to store unsynced events
- Upon reconnection, all queued data was automatically synchronized with Firebase

This ensured zero data loss and maintained continuous tracking regardless of network availability

4. Backend System for Data Analytics and AI Integration

The system used Firebase Realtime Database for structured data storage and real-time updates. The database stored:

- Device status and sensor values
- Dose history records
- Pill count and adherence metrics

The real-time database allowed seamless communication between hardware and mobile application, ensuring immediate data availability

8.2. FRONTEND

1. Mobile Application Interface

A Flutter-based mobile application was developed to provide users with a real-time interface for monitoring medication adherence. The application successfully:

- Displayed current pill count and total doses
- Showed last dose timestamp
- Indicated device status (online/offline)
- Displayed missed dose alerts
- Visualized sensor data and system state

The user interface was designed to be clean, responsive, and user-friendly for effective monitoring.

8.2 IMPLEMENTATION (FRONTEND)

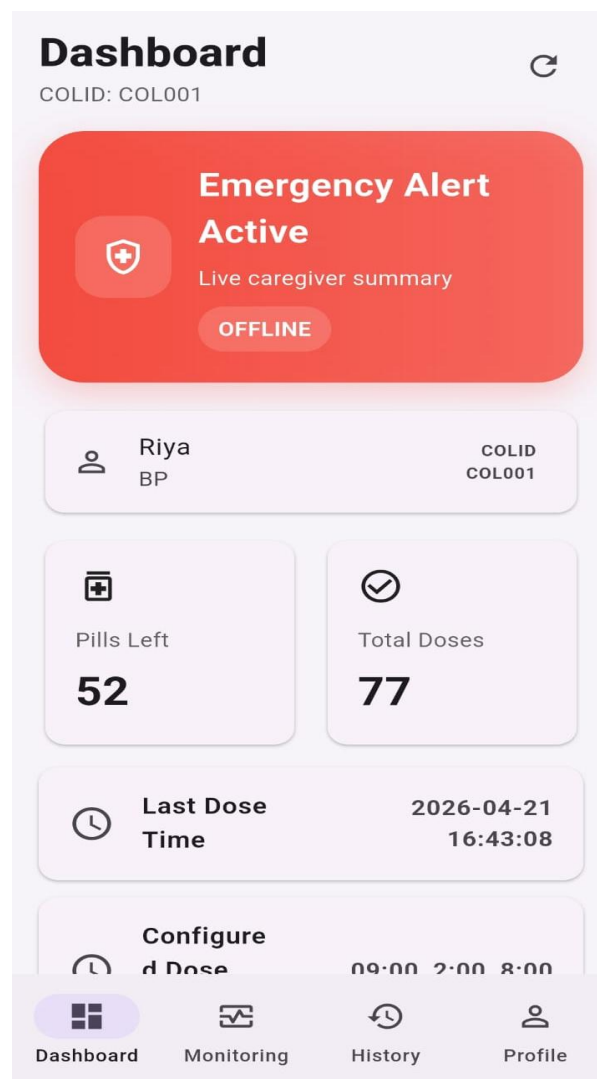


Fig 8.2 UI

2. Real-Time Data Visualization

The application used Firebase listeners to display live updates. Any change in device data was immediately reflected in the app without manual refresh.

This provided:

- Instant feedback to users
- Continuous monitoring capability
- Improved reliability of medication tracking

3. System Integration and Performance

The complete system demonstrated successful integration of:

- Embedded hardware (ESP32 + sensors)
- Cloud backend (Firebase)
- Mobile application (Flutter)

The system operated as a closed-loop solution where:

- Sensors detect dose events
- ESP32 processes logic
- Data is sent to cloud
- App displays real-time updates

The integration ensured efficient, reliable, and autonomous system performance.

8.3 Final Outcome

The implemented system achieved:

- Accurate dose detection using dual-sensor fusion
- Reliable cloud synchronization
- Offline data handling with automatic recovery
- Real-time monitoring via mobile application
- Alert system for missed medication

The project demonstrates a scalable and cost-effective solution for improving medication adherence using IoT technology.

CHAPTER 9 RESULT AND DISCUSSION

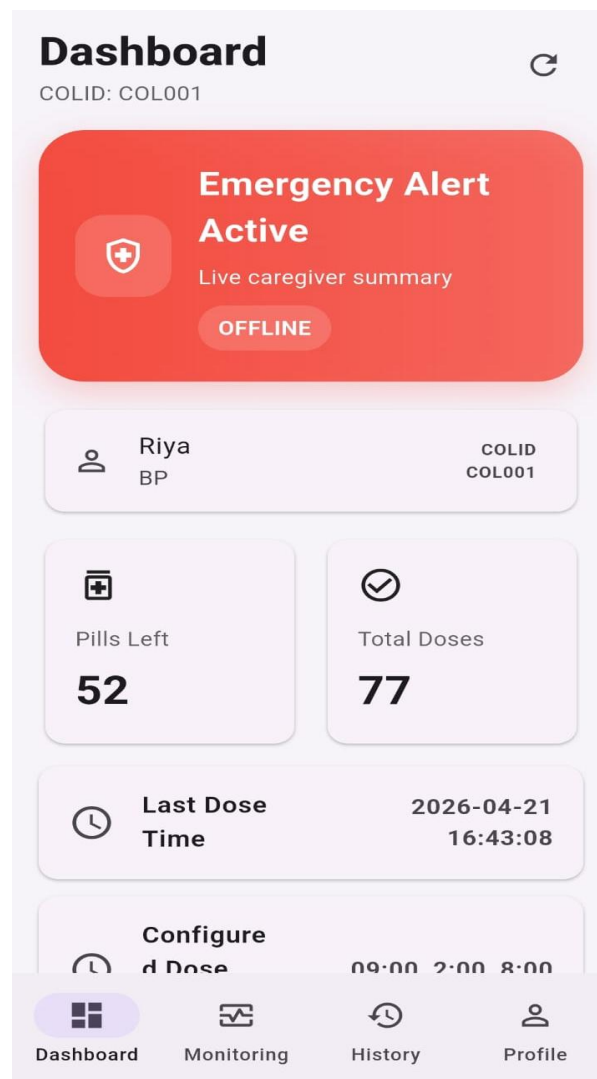


Fig 9.1 Android app showing the dashboard

The developed Flutter-based mobile application successfully displayed real-time data received from the Firebase Realtime Database. The dashboard provided key information including:

- Current pill count
- Total doses taken
- Last dose timestamp
- Device status (online/offline)
- Missed dose alerts

The application updated automatically using real-time Firebase listeners, ensuring that users received instant feedback without manual refresh.

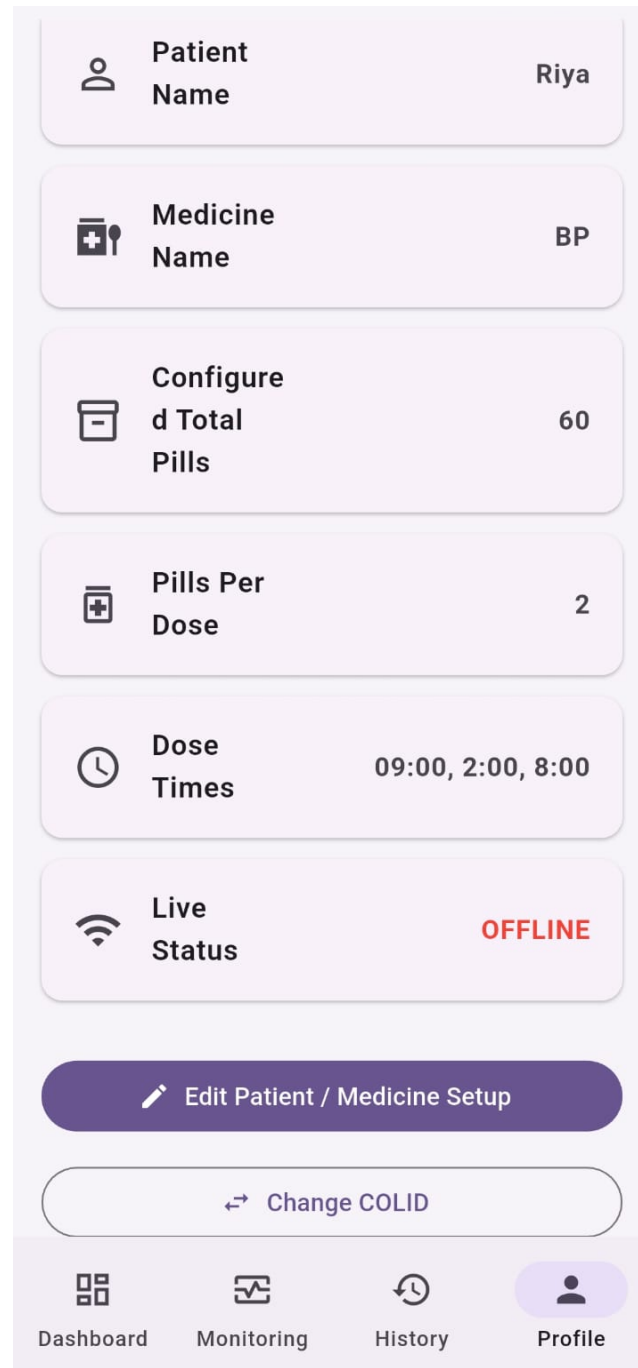


Fig 9.2 Live Interactive dashboard

The system provided continuous monitoring of device activity. The dashboard displayed:

- Cap status (open/closed)
- Movement detection
- Dose confirmation state
- Offline queue count

This allowed users to verify whether a dose was properly taken and whether the system was functioning correctly

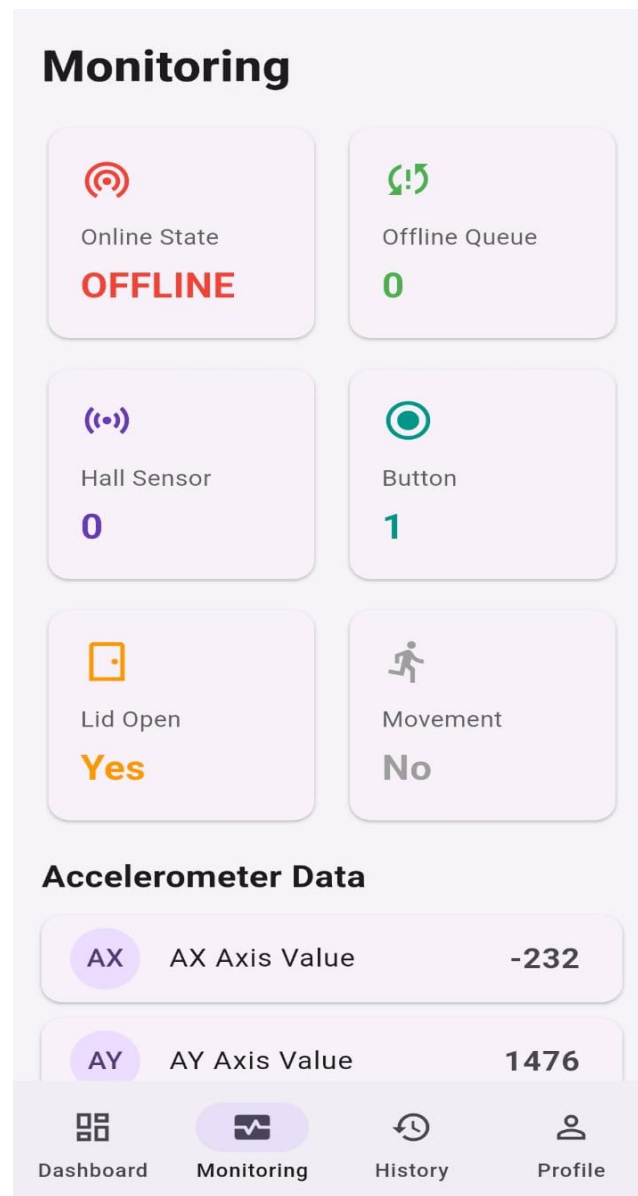


Fig 9.3: Sensor data representation

The MPU6050 accelerometer data was continuously captured and displayed. The variation in sensor values confirmed that the system could accurately detect movement during pill dispensing.

- The X-axis represents time progression
- The Y-axis represents sensor values

The graph demonstrated that significant changes in values corresponded to actual user interaction with the device.

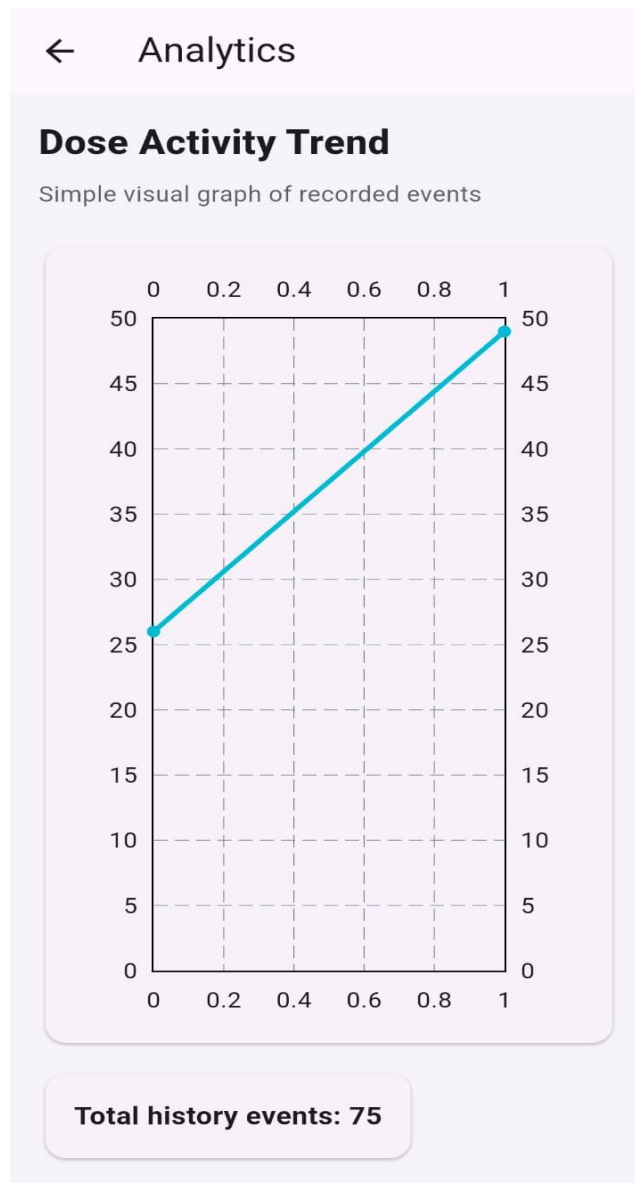


Fig 9.4 Dose detection behavior

The system successfully detected medication intake using a dual-sensor fusion mechanism. A dose was registered only when:

- The lid was opened (Hall sensor triggered)
- Movement was detected (MPU6050 threshold crossed)
- The lid was closed again

This ensured high accuracy and eliminated false detection caused by minor vibrations.

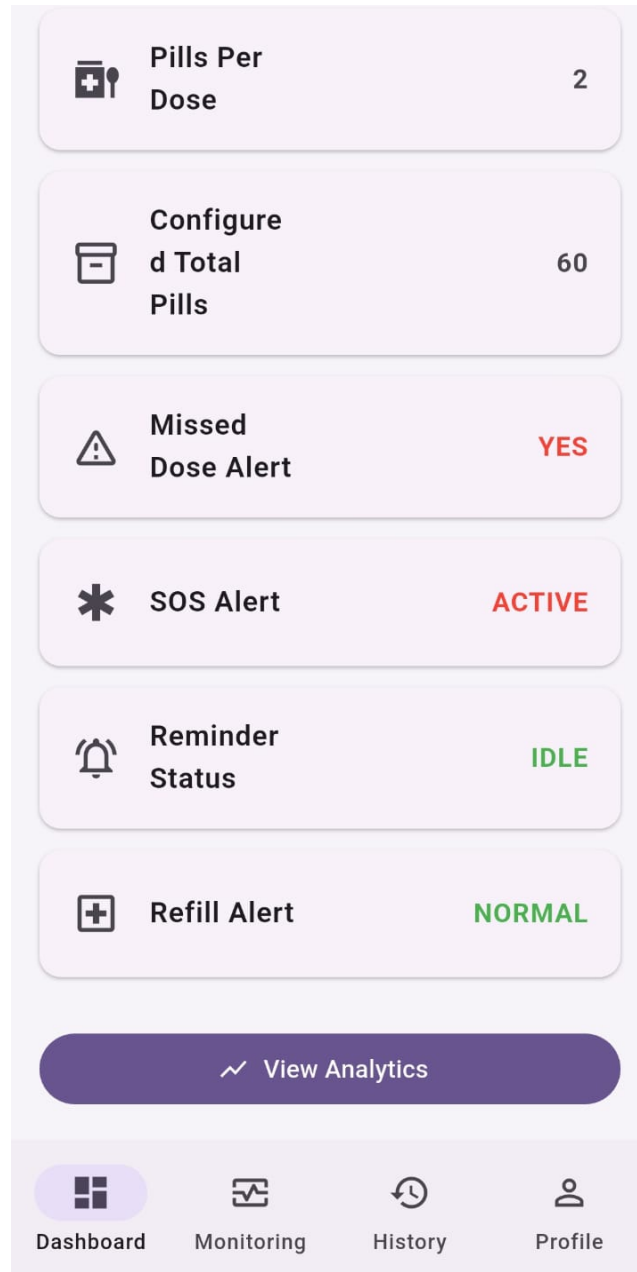


Fig 9.5

The system was tested under offline conditions by disabling Wi-Fi connectivity. During this phase:

- Dose events were stored locally in ESP32 memory
- The offline queue count increased accordingly
- No data loss occurred

When Wi-Fi was restored:

- All queued events were automatically synchronized with Firebase
- The offline queue count returned to zero

This confirmed reliable offline functionality and data recovery.

The system demonstrated stable operation under continuous use. The following observations were made:

- The ESP32 maintained consistent communication with Firebase
- The OLED display provided real-time feedback
- Alerts (buzzer and LED) were triggered correctly during missed doses
- Sensor readings remained consistent and responsive

Minor instability observed in the OLED display during early testing was resolved through improved wiring and optimized update frequency.

DISCUSSION

The developed system effectively integrates embedded hardware, cloud services, and mobile application to provide a complete medication tracking solution. The use of dual-sensor fusion significantly improves detection accuracy compared to single-sensor systems.

The implementation of offline data storage ensures uninterrupted functionality even in the absence of internet connectivity, making the system highly reliable in real-world conditions.

The real-time dashboard enhances usability by providing clear and immediate insights into medication adherence.

FINAL OUTCOME

The experimental results confirm that the system:

- Accurately detects medication intake
- Maintains reliable real-time communication with the cloud
- Handles offline scenarios without data loss
- Provides an intuitive and responsive mobile interface

The system demonstrates strong potential for practical deployment in healthcare environments to improve patient adherence and monitoring.

CHAPTER 10 CONCLUSION & FUTURE WORK

The Smart Medication Adherence Collar was designed and developed as a compact, wearable IoT-based solution to address the growing problem of medication non-adherence among patients with chronic illnesses, elderly individuals, and those with memory impairments. Non-adherence to prescribed medication schedules is one of the leading causes of treatment failure, unnecessary hospitalizations, and increased healthcare costs worldwide. The proposed system directly tackles this challenge by automating the entire medication reminder and tracking process through an integrated hardware and software architecture.

The system successfully integrates multiple hardware components including the ESP32-C3 Supermini microcontroller, A3144 Hall Effect Sensor, MPU6050 motion sensor, OLED display, RGB LED, 3.3V buzzer, tactile button, and a LiPo battery into a single unified wearable collar device. Each component plays a specific and essential role in ensuring that the system operates reliably and continuously throughout the day without requiring constant user intervention.

The collar effectively monitors medication schedules configured via the mobile application, triggers timely audio and visual reminders through the buzzer and RGB LED when medication time is reached, and detects bottle cap opening using the A3144 Hall Effect Sensor to confirm dose intake. Every dose event is logged with an accurate timestamp provided by the RTC module and synced wirelessly to the cloud platform via the ESP32-C3 Supermini's built-in Wi-Fi. This data is securely stored using encryption protocols and compiled into structured compliance reports that are automatically shared with the patient's doctor or caregiver through the mobile application in real time.

Beyond medication tracking, the system also incorporates fall detection and continuous motion monitoring through the MPU6050 IMU sensor, adding an important additional layer of patient safety. If a fall is detected or unusual motion patterns are identified, the system immediately triggers an emergency alert to the caregiver, ensuring that the patient receives timely assistance. The OLED display provides the patient with a clear local interface showing the current time, medication schedule, and dose status without needing to refer to the mobile application at all times.

The closed-loop design of the system ensures that missed doses are immediately detected and caregiver alerts are sent automatically through push notifications or SMS without any manual intervention. The use of the TP4056 charging module with USB Type-C ensures safe and convenient recharging of the LiPo battery, while the Velcro straps provide a comfortable and adjustable fit for patients of different sizes. The overall system is low cost, lightweight, energy efficient, and easy to use, making it accessible and practical for a wide range of patients in both home and clinical environments.

In conclusion, the Smart Medication Adherence Collar successfully demonstrates that a fully automated, sensor-driven, cloud-connected wearable device can significantly improve medication compliance, reduce missed doses, enhance patient safety, and enable proactive healthcare management. The system bridges the gap between the patient and the healthcare provider through real-time data communication and automated reporting, ultimately contributing to better treatment outcomes and improved quality of life for patients who depend on regular medication.

Future Work

The Smart Medication Adherence Collar in its current form establishes a strong and reliable foundation for automated medication tracking and patient safety monitoring. However several promising directions for future research and development have been identified that can further enhance the system's capabilities, accuracy, user experience, and clinical impact.

1. AI-Based Anomaly Detection A machine learning model such as an Isolation Forest or LSTM neural network will be integrated into the cloud platform to analyze historical dose logs and motion patterns, automatically detecting abnormal behavior and sending priority alerts to caregivers before health conditions worsen.

2. Heart Rate and SpO2 Monitoring Future versions of the collar will incorporate a pulse oximetry sensor such as the MAX30102 to monitor the patient's heart rate and blood oxygen levels alongside medication tracking, providing a more comprehensive real-time health monitoring solution.

3. GPS Location Tracking A compact GPS module will be added to track the real-time location of the patient, which is particularly useful for elderly patients with dementia or Alzheimer's who may wander away from home.

4. Voice Assistant Integration A voice-based reminder system will be introduced to assist visually impaired or elderly patients who may have difficulty reading the OLED display, providing spoken medication reminders and confirmations through a small speaker.

5. Multi-Medication Support The current system supports a single medication bottle. Future work will expand this to support multiple medicine bottles simultaneously, each with its own independent schedule, sensor, and dose logging capability.

6. Longer Battery Life Research into energy harvesting techniques such as body heat or solar charging will be explored to extend the battery life of the collar, reducing the frequency of recharging and ensuring uninterrupted operation throughout the day.

7. Doctor Dashboard Web Application A dedicated web-based dashboard will be developed for healthcare providers to monitor multiple patients simultaneously, view detailed compliance reports, set medication schedules remotely, and receive real-time alerts from a single centralized platform.

8. Clinical Trials and Real-World Testing The system will be tested in real clinical environments with actual patients under medical supervision to evaluate its effectiveness, user comfort, reliability, and impact on long-term medication adherence rates.

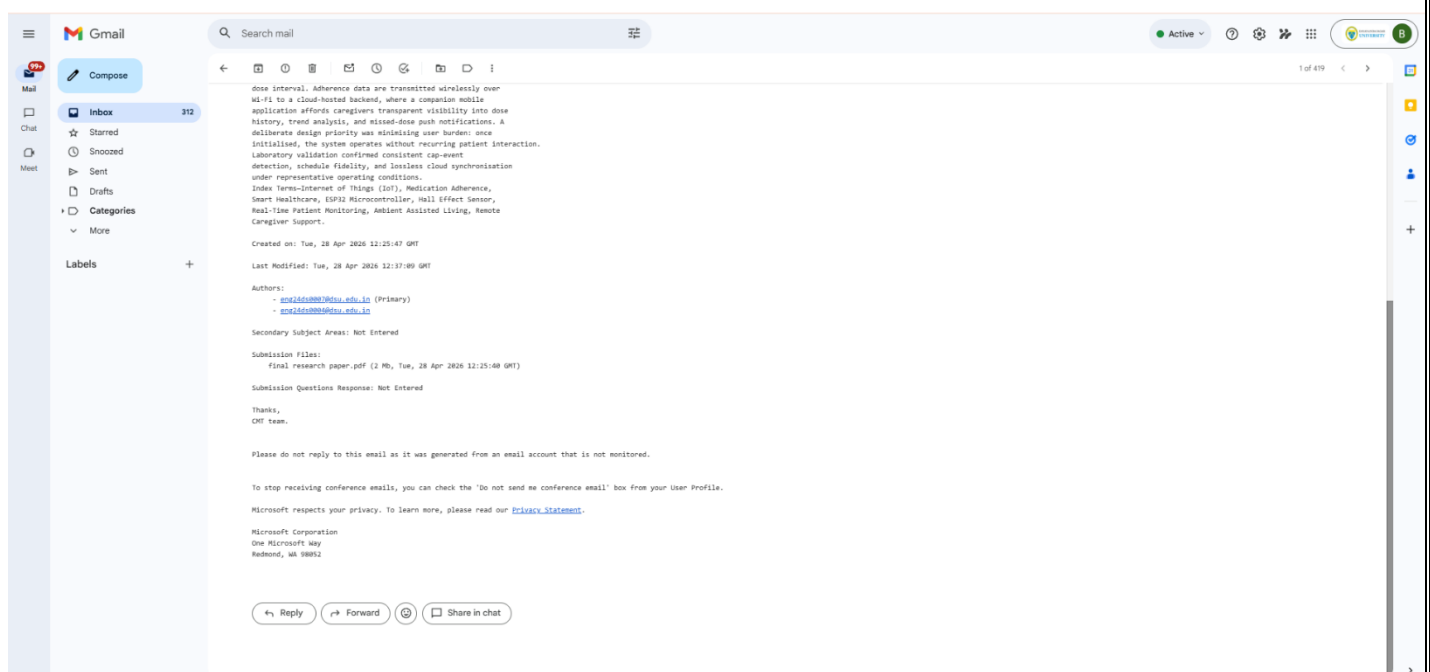
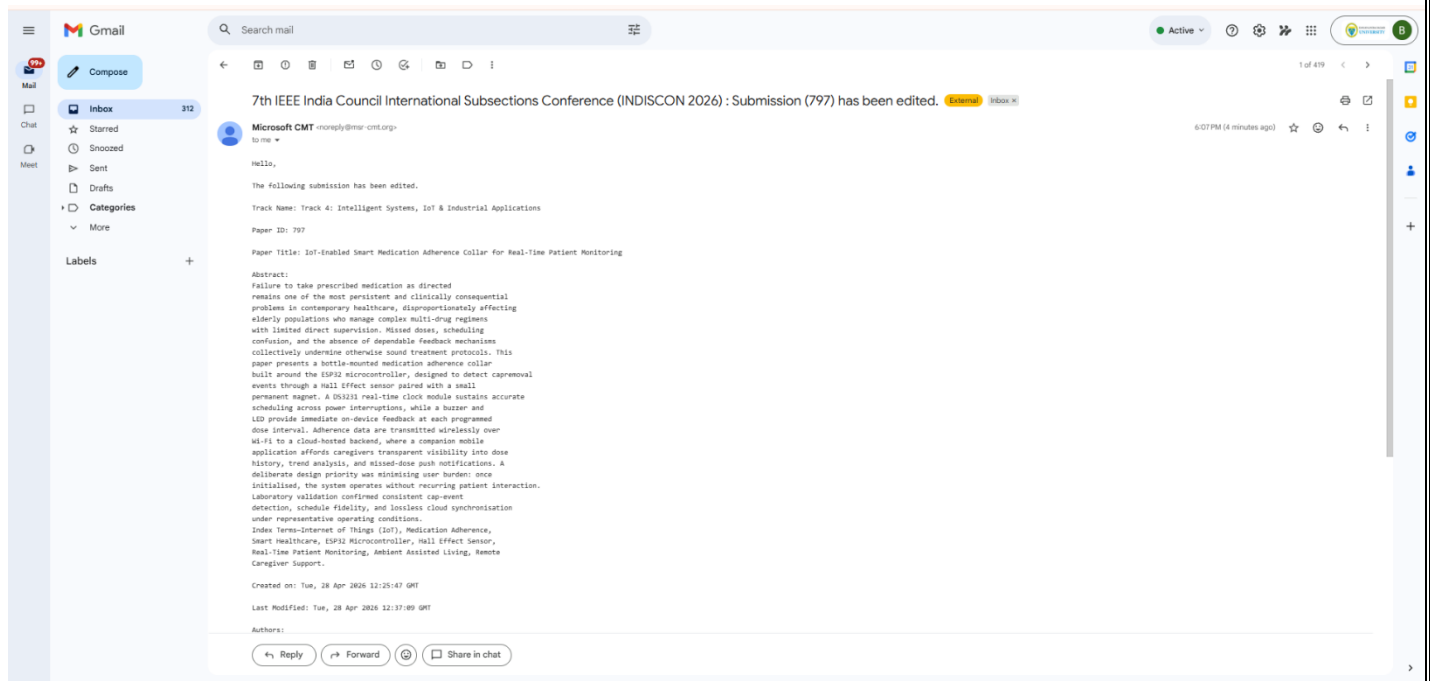
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PUBLISHING DETAILS



PLAGARISM REPORT

